

WASP-2b

R-0088

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_180514 · created 2026-07-15T22:02:50+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458252.9062 +/- 0.0042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.137 +/- 0.018
Transit depth (Rp/Rs)^2	1.88 +/- 0.51 [%]
Orbital Inclination (inc)	86.74 +/- 1.64
Airmass coefficient 1 (a1)	1.012 +/- 0.023
Airmass coefficient 2 (a2)	-0.0077 +/- 0.0085
Scatter in the residuals of the lightcurve fit is	2.21 %
Best Comparison Star	#3 - [529, 119]
Optimal Aperture	4.75
Optimal Annulus	12.89
Transit Duration (day)	0.086 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.137 ± 0.018 vs 0.1326 (deviation 0.2σ)
Duration fit vs published	0.086 d vs 0.0737 d (deviation 0.93σ)
Mid-time O–C	-12.5 min
Depth SNR	6.3
Residual scatter	2.21 %

Light curve

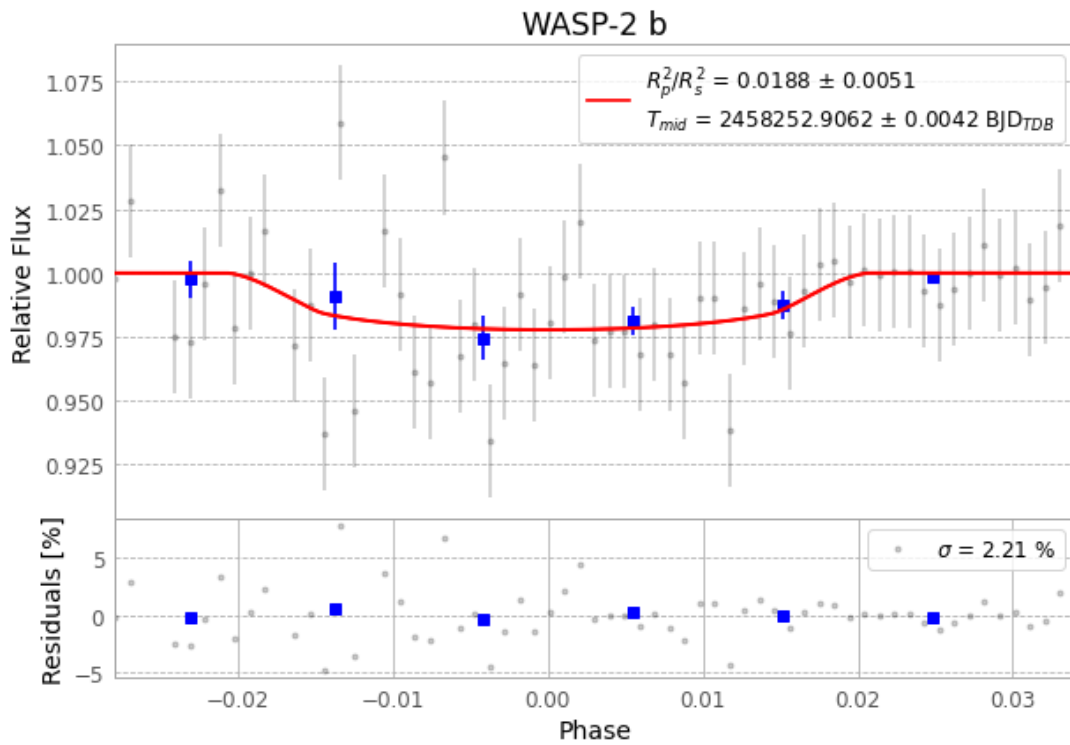


Figure 1 — FinalLightCurve_WASP-2 b_2018-05-14.png (SHA-256 25be0be95f163bde...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [336, 181], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c168b5f614bdea58...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

65 FITS frames (43 MB), WASP-2180514081512.FITS → WASP-2180514112712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-180514.log (5246d0030f83...), FinalLightCurve_WASP-2 b_2018-05-14.png (25be0be95f16...), FinalLightCurve_WASP-2 b_2018-05-14.csv (f93f975f14d7...)

Compute resources

Wall time 215.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 22:02Z.